

of a fund, price of a futures contract, and so on, and then use the same `Correl()` function, which will produce nonsense data. The problem is not the aforementioned Excel function, which is perfectly usable for this purpose, but that you need to have the right input data.

The most common way of calculating correlations is by doing it on log returns, but percentage return numbers will do as well. What you are looking for is the correlation between the changes, not in the absolute values. That's why you cannot use the prices themselves, or even dollar changes, but percentages or log returns.

To calculate the log returns, use the formula:

$$R_i = \text{Ln} \left(\frac{P_i}{P_{i-1}} \right)$$

where P_i is the price or value of your time series at point i in time, P_{i-1} would then be the price of the same time series one day earlier and the resulting R_i is your natural log return for data point i . If you are using Excel to do this, just put all your time series price data in a row and make a row next to it where you use the built-in `Ln()` function to calculate a new column with all the log returns. Then use the `Correl()` function on two sets of log returns and you are done.

Of course, if you're ready to step into a more modern world, you may want to ditch Excel and move straight to Python. This programming language is easy to learn, easy to code, and purpose-built for time series analysis just like what we are doing here.

The correlation coefficient itself will have a value between -1 and 1 , which describes the nature of the relationship between the two series being compared. A value of 1 means that they are identical and move exactly the same, while -1 implies that they move exactly inversely to each other. Zero means that they are completely unrelated.

The correlation matrix in Table 4.3 demonstrates a few interesting points. First, even after the two naïve futures strategies decoupled somewhat after 2013, the overall long-term correlation between them remains fairly high. They were extremely close to each other before, and in the period from 1990 to 2012, they were almost identical.

More interesting is to see how both futures strategies and futures funds have a very low or negative correlation to the stock markets. This is what we are looking for, this is where the real value is. This means that they would